

Anesthesia for Laparoscopy MCQ – Chapter 32

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 32)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Why have laparoscopic surgeries been on the rise over a decade?

- A. Due to better cosmesis only
- B. Due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain
- C. Due to lower cost only
- D. Due to faster anesthesia

Ans: B

Q2. Which gas is most frequently used for creating pneumoperitoneum?

- A. Helium
- B. Carbon dioxide
- C. Argon
- D. Nitrogen

Ans: B

Q3. Which gas is considered the most ideal for laparoscopy but not used due to cost?

- A. Nitrogen
- B. Argon
- C. Helium
- D. Nitrous oxide

Ans: B

Q4. What respiratory effect does increased intra-abdominal pressure cause during pneumoperitoneum?

- A. Increased FRC
- B. Improved ventilation
- C. Increased vital capacity
- D. Pushes diaphragm upwards causing basal atelectasis, V/Q mismatch and decreased FRC

Ans: D

Q5. By how much may thoracopulmonary compliance be decreased during pneumoperitoneum?

- A. Up to 50%
- B. Up to 25%
- C. Up to 10%
- D. Up to 75%

Ans: A

Q6. How does increased intra-abdominal pressure affect cardiac output?

- A. Compresses inferior vena cava leading to decreased venous return and cardiac output
- B. No effect
- C. Increases it
- D. Only affects heart rate

Ans: A

Q7. By how much can intra-abdominal pressure of 12-14 mm Hg decrease cardiac output? A. 10-15% B. 5-10% C. 50-60% D. 30-35%	Ans: D
Q8. Why does peripheral vascular resistance (PVR) increase during laparoscopy? A. Due to patient positioning only B. Due to hypothermia C. Due to blood loss D. Due to high pCO₂ and increased catecholamines, vasopressin and prostaglandins	Ans: D
Q9. What type of arrhythmias are more frequent during laparoscopy due to stretching of peritoneum? A. Ventricular tachycardia B. Atrial fibrillation C. Bradyarrhythmias D. Tachyarrhythmias	Ans: C
Q10. What effect does CO₂ have on the central nervous system during laparoscopy? A. CO₂ by producing cerebral vasodilation can increase the intracranial tension B. Causes CNS depression C. No CNS effect D. Decreases intracranial tension	Ans: A
Q11. Why is the risk of aspiration increased during laparoscopy? A. Due to opioid use B. Increased intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable C. Due to fasting violation D. Due to gas insufflation into stomach	Ans: B
Q12. How can carbon dioxide from peritoneal cavity cause pneumothorax? A. Through the chest wall B. Through the diaphragm directly C. Through embryonic remnants between pleural and peritoneal cavity D. Through the esophagus	Ans: C
Q13. What are the diagnostic signs of pneumothorax during laparoscopy? A. Only chest pain B. Only decreased SpO₂ C. Increased airway resistance, decreased SpO₂, decreased air entry, X-ray chest, and subcutaneous emphysema D. Only hypotension	Ans: C

Q14. What is the FIRST step in managing pneumothorax during laparoscopy? A. Immediately stop nitrous oxide B. Adjust ventilator C. Start 100% oxygen D. Insert chest tube	Ans: A
Q15. Should chest tube drainage be used routinely for capnothorax? A. Only with bilateral pneumothorax B. No, reserved only for cases not responding to conservative measures or life-threatening ones C. Yes, always D. Only in pediatric patients	Ans: B
Q16. In how many minutes does capnothorax usually resolve by itself? A. 10-20 minutes B. 5-10 minutes C. 60-90 minutes D. 30-60 minutes	Ans: D
Q17. Where can CO2 enter to cause pneumomediastinum and subcutaneous emphysema? A. Through the trachea B. Through esophageal, vena cava or aortic hiatus into mediastinum and pericardium C. Through the pleura only D. Through the chest wall	Ans: B
Q18. What is the mortality rate of CO2 embolism during laparoscopy? A. 5% B. 30% C. 50% D. 10%	Ans: B
Q19. How does CO2 embolism occur during laparoscopy? A. Through arterial line B. Through pulmonary veins C. Through IV line D. Due to accidental placement of needle into vessel during insufflation	Ans: D
Q20. What can large volume of gas cause if it enters the inferior vena cava? A. Pneumomediastinum B. Subcutaneous emphysema C. Pneumothorax D. Gas lock leading to cardiovascular collapse	Ans: D

Q21. What is the most sensitive test to diagnose CO2 embolism? A. Transesophageal echocardiography B. Capnography C. Pulmonary artery pressure D. Doppler	Ans: A
Q22. What is the capnography response pattern in CO2 embolism? A. No change in ETCO2 B. Biphasic response - initial rise in end tidal CO2 followed by fall C. Steady decrease in ETCO2 D. Steady increase in ETCO2	Ans: B
Q23. What is diagnostic for CO2 embolism when aspirated from central catheter? A. Blood B. Clear fluid C. Gas or foamy blood D. Saline	Ans: C
Q24. What clinical signs indicate CO2 embolism? A. Tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur B. Hypertension and bradycardia C. Only cyanosis D. Only decreased SpO2	Ans: A
Q25. What must the surgeon ascertain before CO2 insufflation to prevent embolism? A. Oxygen saturation B. Patient's heart rate C. Patient's blood pressure D. Position of Veress needle	Ans: D
Q26. What should the rate of insufflation NOT exceed to prevent CO2 embolism? A. 1 L/min B. 2 L/min C. 3 L/min D. 0.5 L/min	Ans: A
Q27. In what position should the patient be placed for curative management of CO2 embolism? A. Supine B. Trendelenburg C. Durant position (left lateral with Trendelenburg) D. Reverse Trendelenburg	Ans: C

Q28. When may cardiopulmonary bypass be required for CO2 embolism? A. Never required B. Only for pediatric patients C. For massive embolism D. For all cases	Ans: C
Q29. What positions are most laparoscopic procedures performed in? A. Prone and sitting B. Trendelenburg or anti-Trendelenburg C. Supine and lateral D. Lithotomy only	Ans: B
Q30. What abdominal organ trauma can occur during laparoscopy? A. Only splenic injury B. Only bladder injury C. Intestines and stomach may be perforated by needle or trocar with undetected bleeding and visceral trauma D. Only liver injury	Ans: C
Q31. Why can increase in pCO2 during laparoscopy be dangerous in cardiac patients? A. It causes hypotension only B. It has no effect on cardiac patients C. It causes hypothermia D. It can produce arrhythmias and ischemia in cardiac patients	Ans: D
Q32. At what intra-abdominal pressure can cardiac output decrease by more than two-thirds? A. 15 mm Hg B. 12-14 mm Hg C. 10 mm Hg D. > 20 mm Hg	Ans: D
Q33. What is the recommended range for intra-abdominal pressure during laparoscopy? A. 20-25 mm Hg B. 12-14 mm Hg C. 15-20 mm Hg D. 8-10 mm Hg	Ans: B
Q34. What effect can intra-abdominal pressure > 20 mm Hg have on abdominal organs? A. Significantly impair renal, mesenteric and intestinal blood flow B. Only affects kidneys C. Only affects liver D. No effect	Ans: A

Q35. What is the most common postoperative complication of laparoscopic surgeries? A. Infection B. Bleeding C. Nausea and vomiting (incidence 40-75%) D. Pain	Ans: C
Q36. What causes shoulder pain in the postoperative period after laparoscopy? A. Positioning on table B. Surgical retraction C. IV line placement D. Sub-diaphragmatic CO2 irritating diaphragm	Ans: D
Q37. Under what type of anesthesia can very short duration laparoscopic surgeries (<15 minutes) be performed? A. Local anesthesia B. Regional anesthesia C. General anesthesia D. Spinal anesthesia	Ans: A
Q38. Why does CO2 limit the use of central neuraxial blocks for laparoscopy? A. Due to nausea B. Due to increased intracranial tension C. Due to diaphragmatic irritation by CO2 causing patient discomfort (shoulder pain) D. Due to hypotension risk	Ans: C
Q39. What is the most commonly used technique for most laparoscopic surgeries? A. Epidural anesthesia B. Spinal anesthesia with sedation C. Local anesthesia D. General anesthesia with endotracheal intubation and mild hyperventilation	Ans: D
Q40. When can second generation LMA (I-gel, Proseal) be considered for laparoscopy? A. Only in emergency B. Never C. If there is no risk of aspiration D. Always	Ans: C
Q41. At what level should ETCO2 be maintained during laparoscopy? A. 35 mm Hg B. 30 mm Hg C. 45 mm Hg D. 25 mm Hg	Ans: A

Q42. Why should inhalational agents that depress cardiac function (like halothane) be avoided? A. They cause bronchospasm B. Because cardiac output is already compromised by pneumoperitoneum C. They are expensive D. They cause nausea	Ans: B
Q43. Why is propofol the induction agent of choice for laparoscopy? A. Because of its antiemetic properties B. It has fastest onset C. It is cheapest D. It has longest duration	Ans: A
Q44. Which of the following is an ABSOLUTE contraindication for laparoscopy? A. Pregnancy B. Controlled coronary insufficiency C. Congestive heart failure D. Obesity	Ans: C
Q45. Why is right-to-left cardiac shunt an absolute contraindication for laparoscopy? A. Due to arrhythmia B. Risk of paradoxical embolism C. Due to bleeding risk D. Due to hypotension	Ans: B
Q46. What is the relative contraindication for laparoscopy? A. Coronary insufficiency B. Obesity C. Pregnancy D. Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt	Ans: D
Q47. Is laparoscopy considered safe during pregnancy? A. Yes, laparoscopy is considered safe during pregnancy B. Absolutely contraindicated C. Only in first trimester D. Only in third trimester	Ans: A
Q48. Is laparoscopy now preferred over open surgery in obese patients? A. It is equally preferred B. Yes, it has become the preferred method over open surgeries C. Only for minor procedures D. No, it is contraindicated	Ans: B

Q49. Can a patient with controlled ischemic heart disease be safely considered for laparoscopy?

- A. Only with CABG first
- B. Yes, with enhanced perioperative cardiovascular monitoring
- C. Only for short procedures
- D. Absolutely contraindicated

Ans: B

Q50. Are respiratory complications after laparoscopic surgery higher or lower than open surgery?

- A. Significantly lower
- B. Only slightly lower
- C. Same
- D. Higher

Ans: A

Answer Key & Explanations

Q1. Answer: B — Due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain

Page 248: Laparoscopic surgeries have been on rise over a decade due to shorter hospital stay, avoidance of big surgical incision, less tissue trauma and less postoperative pain.

Topic: General

Q2. Answer: B — Carbon dioxide

Page 248: Carbon dioxide is most frequently used gas for laparoscopy because it is more soluble in blood leading to its rapid elimination from the body in cases of pneumothorax and accidental injection into vascular compartment.

Topic: Gases

Q3. Answer: B — Argon

Page 248: Since argon is an inert gas with no side effects it is considered as most ideal gas for laparoscopy, however because of its very high cost it is not used.

Topic: Gases

Q4. Answer: D — Pushes diaphragm upwards causing basal atelectasis, V/Q mismatch and decreased FRC

Page 248: Increase intra-abdominal pressure due to pneumoperitoneum pushes the diaphragm upwards, leading to basal atelectasis producing ventilation perfusion (V/Q) mismatch and decrease in functional residual capacity (FRC).

Topic: Pathophysiology - Respiratory

Q5. Answer: A — Up to 50%

Page 248: Thoracopulmonary compliance may be decreased by up to 50% during pneumoperitoneum.

Topic: Pathophysiology - Respiratory

Q6. Answer: A — Compresses inferior vena cava leading to decreased venous return and cardiac output

Page 248: Increase intra-abdominal pressure compresses the inferior vena cava leading to decreased venous return and cardiac output.

Topic: Pathophysiology - Cardiovascular

Q7. Answer: D — 30-35%

Page 248: Intra-abdominal pressure of 12-14 mm Hg can decrease the cardiac output by 30-35%.

Topic: Pathophysiology - Cardiovascular

Q8. Answer: D — Due to high pCO₂ and increased catecholamines, vasopressin and prostaglandins

Page 249: Increase in PVR occurs because of high pCO₂, and more importantly, because of increased catecholamines, vasopressin and prostaglandins.

Topic: Pathophysiology - Cardiovascular

Q9. Answer: C — Bradyarrhythmias

Page 249: CO₂ by producing coronary vasoconstriction may produce any arrhythmia however, bradyarrhythmias occurring because of stretching of the peritoneum are more frequent.

Topic: Pathophysiology - Cardiovascular

Q10. Answer: A — CO2 by producing cerebral vasodilation can increase the intracranial tension

Page 248: CO2 by producing cerebral vasodilation can increase the intracranial tension.

Topic: Pathophysiology - CNS

Q11. Answer: B — Increased intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable

Page 248: Increased Intra-abdominal pressure distorts the gastroesophageal junction making patient more vulnerable for aspiration.

Topic: Pathophysiology - GI

Q12. Answer: C — Through embryonic remnants between pleural and peritoneal cavity

Page 249: Carbon dioxide from peritoneal cavity can diffuse into the pleural cavity through embryonic remnants present between pleural and peritoneal cavity which gets open by increased intra-abdominal pressure or through pleural tears during surgery.

Topic: Complications - Pneumothorax

Q13. Answer: C — Increased airway resistance, decreased SpO2, decreased air entry, X-ray chest, and subcutaneous emphysema

Page 249: Diagnosis includes increased airway resistance on mechanical ventilation, decreased oxygen saturation (SpO2), decreased air entry on auscultation, X-ray chest, abnormal movement of diaphragm, and associated subcutaneous emphysema of neck and chest.

Topic: Complications - Pneumothorax

Q14. Answer: A — Immediately stop nitrous oxide

Page 249: Management - Immediately stop nitrous oxide (it can expand the pneumothorax).

Topic: Complications - Pneumothorax

Q15. Answer: B — No, reserved only for cases not responding to conservative measures or life-threatening ones

Page 249: Chest tube drainage should be reserved only for the cases not responding to conservative measures or the Pneumothorax is life-threatening.

Topic: Complications - Pneumothorax

Q16. Answer: D — 30-60 minutes

Page 249: Usually capnothorax resolves by itself in 30-60 minutes (because of high diffusibility of CO2).

Topic: Complications - Pneumothorax

Q17. Answer: B — Through esophageal, vena cava or aortic hiatus into mediastinum and pericardium

Page 249: The CO2 can enter mediastinum and pericardium through esophageal, vena cava or aortic hiatus. Gas from mediastinum may diffuse cephalad to produce subcutaneous emphysema of face and neck.

Topic: Complications - Pneumomediastinum

Q18. Answer: B — 30%

Page 249: Gas embolism can be a fatal complication of laparoscopy; it carries a mortality of 30%.

Topic: Complications - CO2 Embolism

Q19. Answer: D — Due to accidental placement of needle into vessel during insufflation

Page 249: CO2 embolism occurs because of accidental placement of a needle into the vessel during insufflation.

Topic: Complications - CO2 Embolism

Q20. Answer: D — Gas lock leading to cardiovascular collapse

Page 249: Large volume of gas can cause 'gas lock' in the inferior vena cava leading to cardiovascular collapse.

Topic: Complications - CO2 Embolism

Q21. Answer: A — Transesophageal echocardiography

Page 249: Most sensitive test to diagnose CO2 embolism is transesophageal echocardiography.

Topic: Complications - CO2 Embolism

Q22. Answer: B — Biphasic response - initial rise in end tidal CO2 followed by fall

Page 249: In CO2 embolism, there is biphasic response, i.e. initial rise in end tidal CO2 (due to direct injection of CO2 in vessels) followed by fall.

Topic: Complications - CO2 Embolism

Q23. Answer: C — Gas or foamy blood

Page 249: Aspiration of gas or foamy blood from central catheter is diagnostic.

Topic: Complications - CO2 Embolism

Q24. Answer: A — Tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur

Page 249: Clinical signs like tachycardia, hypotension, right heart strain on ECG and Mill wheel murmur.

Topic: Complications - CO2 Embolism

Q25. Answer: D — Position of Veress needle

Page 249: The surgeon must ascertain the position of Veress needle before CO2 insufflation.

Topic: Complications - CO2 Embolism

Q26. Answer: A — 1 L/min

Page 249: Rate of insufflation should not be more than 1 L/min.

Topic: Complications - CO2 Embolism

Q27. Answer: C — Durant position (left lateral with Trendelenburg)

Page 249: If possible place the patient in Durant position (left lateral with Trendelenburg) so that gas get remains in right ventricle and can be aspirated.

Topic: Complications - CO2 Embolism

Q28. Answer: C — For massive embolism

Page 249: For massive embolism cardiopulmonary bypass may be required.

Topic: Complications - CO2 Embolism

Q29. Answer: B — Trendelenburg or anti-Trendelenburg

Page 250: Majority of the laparoscopic procedures are performed in either Trendelenburg or anti-Trendelenburg position.

Topic: Complications - Positioning

Q30. Answer: C — Intestines and stomach may be perforated by needle or trocar with undetected bleeding and visceral trauma

Page 250: Abdominal organs, especially intestines and stomach may be perforated by the needle or trocar. Moreover, bleeding and visceral trauma may go undetected.

Topic: Complications - Organ Trauma

Q31. Answer: D — It can produce arrhythmias and ischemia in cardiac patients

Page 250: Increase in pCO₂ is well-tolerated by young individuals, but can produce arrhythmias and ischemia in cardiac patients.

Topic: Complications - pCO₂

Q32. Answer: D — > 20 mm Hg

Page 250: Intra-abdominal pressure of > 20 mm Hg can decrease the cardiac output by more than two-thirds producing even cardiac arrest.

Topic: Complications - IAP

Q33. Answer: B — 12-14 mm Hg

Page 250: It is highly recommended that intra-abdominal pressure should be kept between 12-14 mm Hg.

Topic: Complications - IAP

Q34. Answer: A — Significantly impair renal, mesenteric and intestinal blood flow

Page 250: Intra-abdominal pressure of > 20 mm Hg can significantly impair renal, mesenteric and intestinal blood flow.

Topic: Complications - IAP

Q35. Answer: C — Nausea and vomiting (incidence 40-75%)

Page 250: Nausea and vomiting is the most common postoperative complication of laparoscopic surgeries; incidence can be as high as 40-75%.

Topic: Complications - PONV

Q36. Answer: D — Sub-diaphragmatic CO₂ irritating diaphragm

Page 250: Shoulder pain: Sub-diaphragmatic CO₂ irritating diaphragm causing shoulder pain in postoperative period is commonly complained by many patients.

Topic: Complications - Shoulder Pain

Q37. Answer: A — Local anesthesia

Page 250: Laparoscopic surgeries of very short duration (<15 minutes) can only be considered under local anesthesia. Laparoscopic tubal ligation is the most common surgery performed under local anesthesia.

Topic: Anesthetic Management

Q38. Answer: C — Due to diaphragmatic irritation by CO₂ causing patient discomfort (shoulder pain)

Page 250: Patient discomfort (shoulder pain due to diaphragmatic irritation by CO₂) limits the use of central neuraxial blocks (spinal/epidural) for laparoscopy.

Topic: Anesthetic Management

Q39. Answer: D — General anesthesia with endotracheal intubation and mild hyperventilation

Page 250: General anesthesia with endotracheal intubation and mild hyperventilation is most commonly used technique for most of the laparoscopic surgeries.

Topic: Anesthetic Management

Q40. Answer: C — If there is no risk of aspiration

Page 250: General anesthesia with second generation laryngeal mask airway (I-gel, proseal) can be considered if there is no risk of aspiration.

Topic: Anesthetic Management

Q41. Answer: A — 35 mm Hg

Page 250: ETCO₂ should be maintained at 35 mm Hg.

Topic: Anesthetic Management

Q42. Answer: B — Because cardiac output is already compromised by pneumoperitoneum

Page 250: Inhalational agents which depress cardiac function (like halothane) should be avoided.

Topic: Anesthetic Management

Q43. Answer: A — Because of its antiemetic properties

Page 250: Propofol because of its antiemetic properties is the induction agent of choice.

Topic: Anesthetic Management

Q44. Answer: C — Congestive heart failure

Page 250: Absolute contraindications include: Increased intracranial tension, congestive heart failure, severe hypovolemia and hypotension, right-to-left cardiac shunts or patent foramen ovale, generalized abdominal peritonitis, and retinal detachment.

Topic: Contraindications

Q45. Answer: B — Risk of paradoxical embolism

Page 250: Right-to-left cardiac shunts or patent foramen ovale (risk of paradoxical embolism) is an absolute contraindication.

Topic: Contraindications

Q46. Answer: D — Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt

Page 250: Relative contraindication - Ventriculoperitoneal (VP) or peritoneojugular (PJ) shunt: Insufflation in peritoneum can displace the shunt.

Topic: Contraindications

Q47. Answer: A — Yes, laparoscopy is considered safe during pregnancy

Page 250: Although pneumoperitoneum may cause transient fetal acidosis, clinical studies suggest that additional risks to be very low for both mother and fetus. Therefore, laparoscopy is considered safe during pregnancy.

Topic: Contraindications - Past

Q48. Answer: B — Yes, it has become the preferred method over open surgeries

Page 250: Laparoscopy once used to be considered as contraindication in obese patients has now become the preferred method over open surgeries.

Topic: Contraindications - Past

Q49. Answer: B — Yes, with enhanced perioperative cardiovascular monitoring

Page 250: A patient with the controlled ischemic heart disease can be safely considered for laparoscopic surgeries however, with enhanced perioperative cardiovascular monitoring.

Topic: Contraindications - Past

Q50. Answer: A — Significantly lower

Page 250: Respiratory complications after laparoscopic surgery are significantly lower than open surgery.

Topic: Contraindications - Past